

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
df = pd.read_csv('/content/aerofit_treadmill.txt')
df.head()
```

	Product	Age	Gender	Education	MaritalStatus	Usage	Fitness	Income	Miles
0	KP281	18	Male	14	Single	3	4	29562	112
1	KP281	19	Male	15	Single	2	3	31836	75
2	KP281	19	Female	14	Partnered	4	3	30699	66
3	KP281	19	Male	12	Single	3	3	32973	85
4	KP281	20	Male	13	Partnered	4	2	35247	47

Next steps: [Generate code with df](#) [View recommended plots](#)

1. Import the dataset and do usual data analysis steps like checking the structure & characteristics of the dataset

a. The data type of all columns in the "customers" table.

```
df.dtypes
```

```
Product      object
Age          int64
Gender       object
Education    int64
MaritalStatus object
Usage        int64
Fitness      int64
Income       int64
Miles        int64
dtype: object
```

b. we can find the number of rows and columns given in the dataset

```
df.shape
```

```
(180, 9)
```

c. Check for the missing values and find the number of missing values in each column

```
missing_values = df.isnull().sum()
print(missing_values)
```

```
Product      0
Age          0
Gender       0
Education    0
MaritalStatus 0
```

```

Usage      0
Fitness    0
Income     0
Miles      0
dtype: int64

```

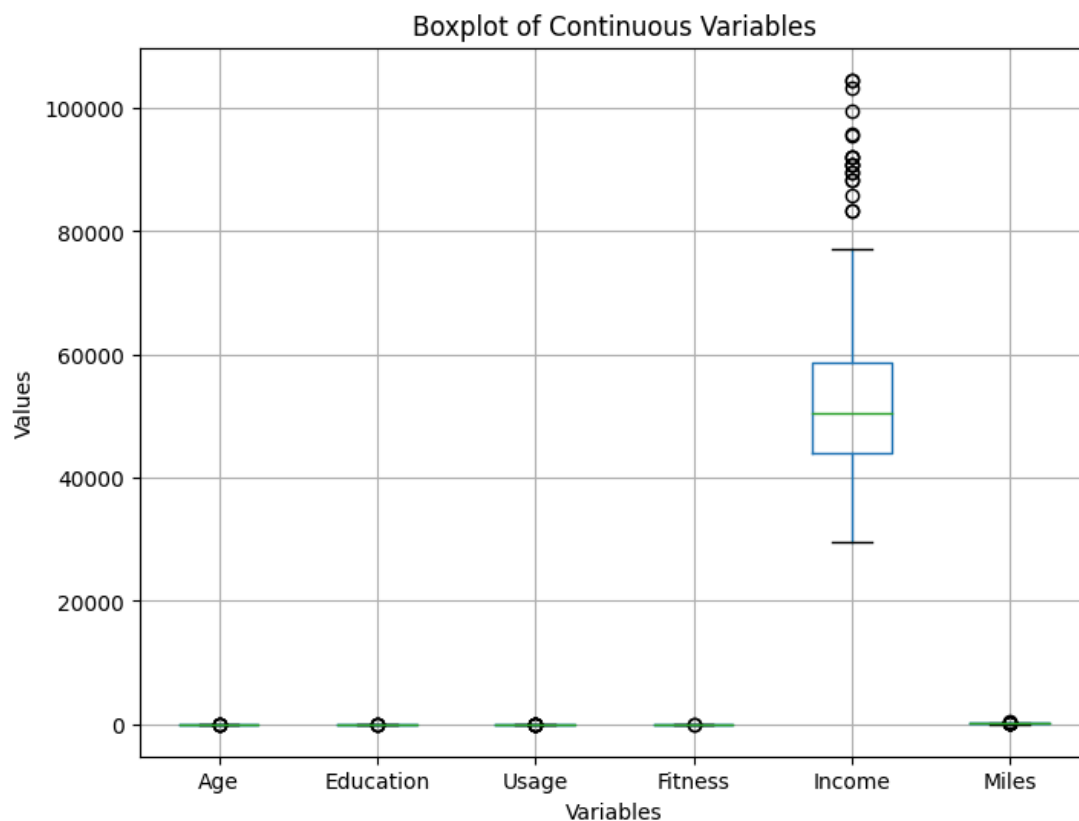
2. Detect Outliers

a. Find the outliers for every continuous variable in the dataset

```

plt.figure(figsize=(8, 6))
df.boxplot()
plt.title('Boxplot of Continuous Variables')
plt.ylabel('Values')
plt.xlabel('Variables')
plt.show()

```



b. Remove/clip the data between the 5 percentile and 95 percentile

```

percentile_5 = df['Income'].quantile(0.05)
percentile_95 = df['Income'].quantile(0.95)
df['Income'] = df['Income'].clip(percentile_5, percentile_95)
df['Income']

```



```

0      34109.857875
1      34109.857875
2      34109.857875
3      34109.857875
4      35247.000000
...

```

```

175      83416.000000
176      89641.000000
177      90886.000000
178      90886.155625
179      90886.155625
Name: Income, Length: 180, dtype: float64

```

3. Check if features like marital status, Gender, and age have any effect on the product purchased

a. Find if there is any relationship between the categorical variables and the output variable in the data.

```

plt.figure(figsize=(20,7)).suptitle('check feature marital status, Gender, Age on the product purchased', fontsize=14)
plt.subplot(1,3,1)
sns.countplot(x='MaritalStatus', hue='Product', data=df)
plt.title('Marital Status vs. Product')
plt.xlabel('MaritalStatus')
plt.ylabel('Count')
plt.legend(title='Product')

plt.subplot(1,3,2)
sns.countplot(x='Gender', hue='Product', data=df)
plt.title('Gender vs. product')
plt.xlabel('Gender')
plt.ylabel('Count')
plt.legend(title='product')

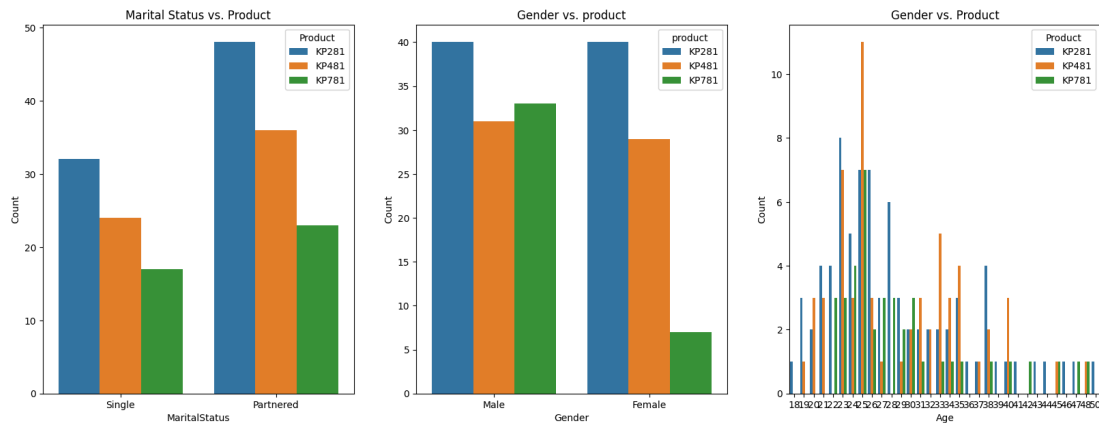
plt.subplot(1,3,3)
sns.countplot(x='Age', hue='Product', data=df)
plt.title('Gender vs. Product')
plt.xlabel('Age')
plt.ylabel('Count')
plt.legend(title='Product')

```



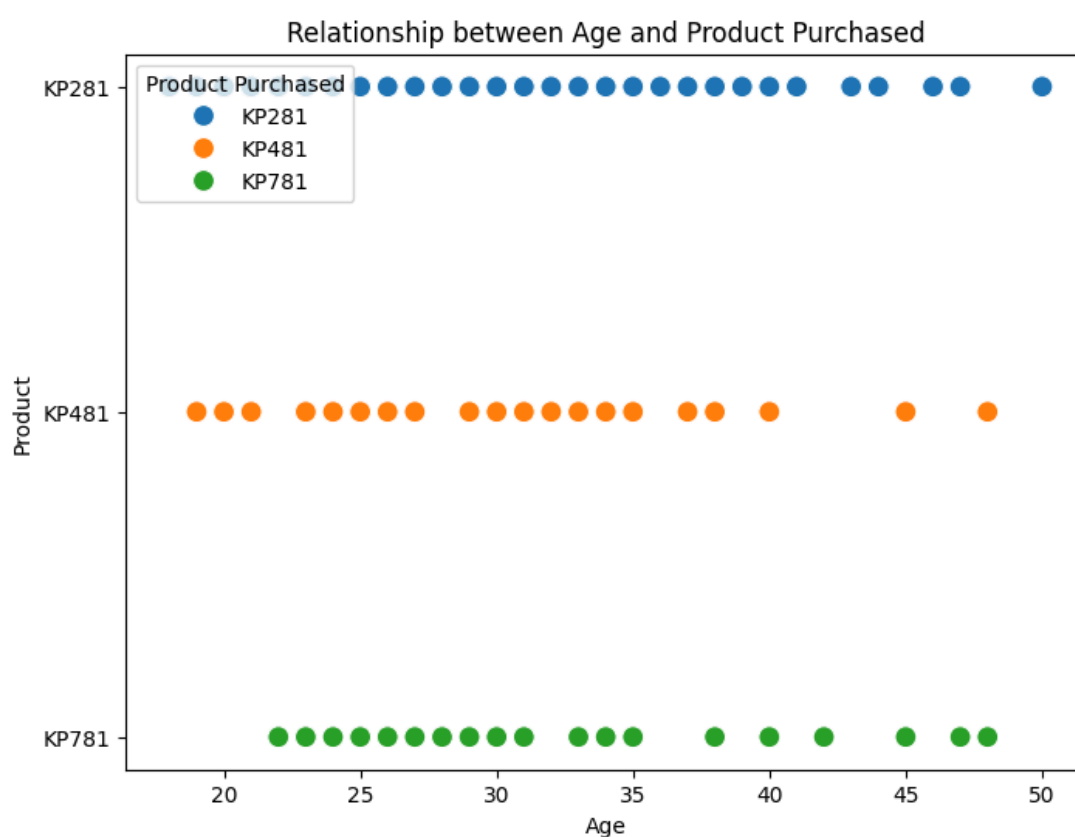
<matplotlib.legend.Legend at 0x7ad5a4217c40>

check feature marital status, Gender, Age on the product purchased



b. Find if there is any relationship between the continuous variables and the output variable in the data.

```
plt.figure(figsize=(8, 6))
sns.scatterplot(x='Age', y='Product', data=df, hue='Product', marker='o', s=100)
plt.title('Relationship between Age and Product Purchased')
plt.xlabel('Age')
plt.ylabel('Product')
plt.legend(title='Product Purchased', loc='upper left')
plt.show()
```



4. Representing the Probability

a. Find the marginal probability (what percent of customers have purchased KP281, KP481, or KP781)

```
contingency_table = pd.crosstab(index=df['Product'], columns='count', normalize=True)
```

```
# Print the marginal probability of each product
print("Marginal probability of each product:")
print(contingency_table)
```

```

Marginal probability of each product:
col_0    count
Product
KP281    0.444444
KP481    0.333333
KP781    0.222222

```

b. Find the probability that the customer buys a product based on each column.

```
conditional_probs = {}

# Iterate over each feature column
for column in df.columns[:-1]: # Exclude the last column (product)
    # Calculate conditional probabilities for each feature value
    conditional_prob = pd.crosstab(index=df['Product'], columns=df[column], normalize='columns')
    conditional_probs[column] = conditional_prob

# Print conditional probabilities for each feature
for feature, conditional_prob in conditional_probs.items():
    print("Conditional probabilities for feature:", feature)
    print(conditional_prob)
    print()
```



[3 rows x 32 columns]

Conditional probabilities for feature: Gender

Gender	Female	Male
Product		
KP281	0.526316	0.384615
KP481	0.381579	0.298077
KP781	0.092105	0.317308

Conditional probabilities for feature: Education

Education	12	13	14	15	16	18	20	21
Product								
KP281	0.666667	0.6	0.545455	0.8	0.458824	0.086957	0.0	0.0
KP481	0.333333	0.4	0.418182	0.2	0.364706	0.086957	0.0	0.0
KP781	0.000000	0.0	0.036364	0.0	0.176471	0.826087	1.0	1.0

Conditional probabilities for feature: MaritalStatus

MaritalStatus	Partnered	Single
Product		
KP281	0.448598	0.438356
KP481	0.336449	0.328767
KP781	0.214953	0.232877

Conditional probabilities for feature: Usage

Usage	2	3	4	5	6	7
Product						
KP281	0.575758	0.536232	0.423077	0.117647	0.0	0.0

```

KP481      0.0      0.0      0.0      0.0
KP781      1.0      1.0      1.0      1.0

```

```
[3 rows x 62 columns]
```

C.Find the conditional probability that an event occurs given that another event has occurred. (Example: given that a customer is female, what is the probability she'll purchase a KP481)

```

joint_prob_female_KP481 = (df['Gender'] == 'Female') & (df['Product'] == 'KP481')
joint_prob_female_KP481 = joint_prob_female_KP481.mean() # Convert to probability by taking the mean

# Calculate probability of being female
prob_female = (df['Gender'] == 'Female').mean() # Convert to probability by taking the mean

# Calculate conditional probability
conditional_prob_KP481_given_female = joint_prob_female_KP481 / prob_female

print("Conditional probability of purchasing KP481 given female:", conditional_prob_KP481_given_female)

```

Conditional probability of purchasing KP481 given female: 0.3815789473684211

5. Check the correlation among different factors

a.Find the correlation between the given features in the table.

```
df.select_dtypes(['int', 'float']).corr()
```



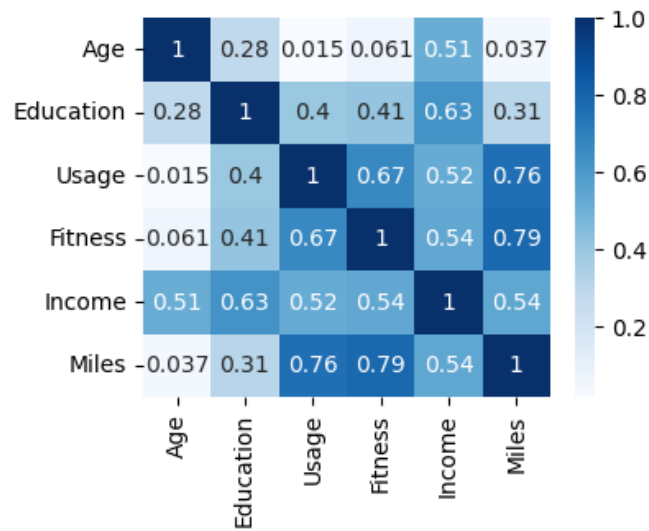
	Age	Education	Usage	Fitness	Income	Miles
Age	1.000000	0.280496	0.015064	0.061105	0.513414	0.036618
Education	0.280496	1.000000	0.395155	0.410581	0.625827	0.307284
Usage	0.015064	0.395155	1.000000	0.668606	0.519537	0.759130
Fitness	0.061105	0.410581	0.668606	1.000000	0.535005	0.785702
Income	0.513414	0.625827	0.519537	0.535005	1.000000	0.543473
Miles	0.036618	0.307284	0.759130	0.785702	0.543473	1.000000



```

plt.figure(figsize=(4, 3))
sns.heatmap(df.select_dtypes(['int', 'float']).corr(), cmap='Blues', annot=True)
plt.show()

```



6. Customer profiling and recommendation

a. Make customer profilings for each and every product. Hint: We want you to find at What age, gender, and income group but product the KP281

```
kp281_data = df[df['Product'] == 'KP281']

# Group by age, gender, and income
grouped_data = kp281_data.groupby(['Age', 'Gender', 'Income']).size().reset_index(name='count')

# Calculate statistics
age_stats = grouped_data['Age'].describe()
gender_stats = grouped_data['Gender'].value_counts()
income_stats = grouped_data['Income'].describe()
plt.figure(figsize=(12, 8))

# Age distribution
plt.subplot(3, 1, 1)
sns.histplot(kp281_data['Age'], bins=20, kde=True)
plt.title('Age Distribution for KP281')
plt.xlabel('Age')

# Gender distribution
plt.subplot(3, 1, 2)
sns.countplot(x='Gender', data=kp281_data)
plt.title('Gender Distribution for KP281')
plt.xlabel('Gender')

# Income distribution
plt.subplot(3, 1, 3)
sns.histplot(kp281_data['Income'], bins=20, kde=True)
plt.title('Income Distribution for KP281')
plt.xlabel('Income')

plt.tight_layout()
plt.show()

# Print statistics
print("Age Statistics for KP281:")
print(age_stats)
print("\nGender Distribution for KP281:")
print(gender_stats)
print("\nIncome Statistics for KP281:")
print(income_stats)
```




Age Distribution for KB281

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10.0 |

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